

MATH-1014 T10A

Tutorial 3

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Today's Roadmap

- 1 Integration by Parts
- 2 Volumes of Revolution
- 3 Mean Value Theorem (MVT)

Review: Integration by Parts

The Formula

$$\int u \, dv = uv - \int v \, du$$

Strategy (LIATE): Prioritize picking u in this order:

- 1 **L**ogarithmic ($\ln x$)
- 2 **I**nverse Trig ($\tan^{-1} x$)
- 3 **A**lgebraic (x^n)
- 4 **T**rigonometric ($\sin x$)
- 5 **E**xponential (e^x)

Problem 1: Inverse Trig Functions

Problem 1. Evaluate:

$$\int_0^1 \tan^{-1} x \, dx$$

Hint: Since there is only one function, choose $u = \tan^{-1} x$ and $dv = dx$.

Solution 1

Setup:

$$\begin{aligned} u = \tan^{-1} x &\implies du = \frac{1}{1+x^2} dx \\ dv = dx &\implies v = x \end{aligned}$$

Apply Formula:

$$I = \left[x \tan^{-1} x \right]_0^1 - \int_0^1 \frac{x}{1+x^2} dx$$

Calculate Integral: For $\int \frac{x}{1+x^2} dx$, let $t = 1 + x^2 \implies dt = 2x dx$.

$$\int_0^1 \frac{x}{1+x^2} dx = \frac{1}{2} \int_1^2 \frac{dt}{t} = \frac{1}{2} [\ln t]_1^2 = \frac{1}{2} \ln 2$$

Final Answer:

$$I = \left(1 \cdot \frac{\pi}{4} - 0 \right) - \frac{1}{2} \ln 2 = \frac{\pi}{4} - \frac{1}{2} \ln 2$$

Problem 2: Cyclic Integration

Problem 2. Evaluate:

$$I = \int e^x \sin x \, dx$$

Strategy

Apply IBP twice to get back to the original integral, then solve algebraically.

Solution 2

Step 1: Let $u = \sin x$, $dv = e^x dx \implies du = \cos x dx$, $v = e^x$.

$$I = e^x \sin x - \int e^x \cos x dx \quad \dots (*1)$$

Step 2: For $\int e^x \cos x dx$, let $U = \cos x$, $dV = e^x dx$.

$$\int e^x \cos x dx = e^x \cos x - \int e^x (-\sin x) dx = e^x \cos x + I$$

Step 3: Substitute back into (*1):

$$I = e^x \sin x - (e^x \cos x + I)$$

$$I = e^x \sin x - e^x \cos x - I$$

$$2I = e^x (\sin x - \cos x)$$

$$I = \frac{1}{2} e^x (\sin x - \cos x) + C$$

Problem 3: Reduction Formula

Problem 3. Use integration by parts to show that for an integer $n \geq 2$:

$$\int_0^{\pi/2} \sin^n x \, dx = \frac{n-1}{n} \int_0^{\pi/2} \sin^{n-2} x \, dx$$

Start by splitting the power:

$$\sin^n x = \sin^{n-1} x \cdot \sin x$$

Solution 3

Let $I_n = \int_0^{\pi/2} \sin^n x \, dx$. Set $u = \sin^{n-1} x$ and $dv = \sin x \, dx$.

$$du = (n-1) \sin^{n-2} x \cos x \, dx, \quad v = -\cos x$$

1. Apply IBP:

$$\begin{aligned} I_n &= \left[-\sin^{n-1} x \cos x \right]_0^{\pi/2} - \int_0^{\pi/2} (-\cos x)(n-1) \sin^{n-2} x \cos x \, dx \\ &= (0 - 0) + (n-1) \int_0^{\pi/2} \sin^{n-2} x \cos^2 x \, dx \end{aligned}$$

2. Use Identity $\cos^2 x = 1 - \sin^2 x$:

$$I_n = (n-1) \int_0^{\pi/2} \sin^{n-2} x (1 - \sin^2 x) \, dx = (n-1)I_{n-2} - (n-1)I_n$$

3. Rearrange:

$$I_n + (n-1)I_n = (n-1)I_{n-2} \implies nI_n = (n-1)I_{n-2} \implies \mathbf{I_n = \frac{n-1}{n} I_{n-2}}$$

Problem 4: Comparison of Methods

Problem 4. Let $\mathcal{R}$ be the region bounded by $y = \sqrt{x}$ and $y = x^2$. Find the volume of the solid generated by rotating $\mathcal{R}$ about the line $x = -1$.

Analyze:

- **Intersections:**

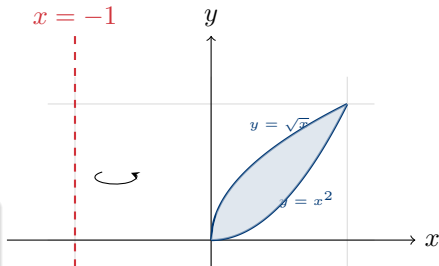
$$\sqrt{x} = x^2 \implies x = 0, 1.$$

- **Axis of Rotation:** $x = -1$ (Offset from origin!).

- **Region:** $y = \sqrt{x}$ is Above, $y = x^2$ is Below on $(0, 1)$.

Challenge

- **Shells:** Vertical strips (dx). Radius changes.
- **Washers:** Horizontal strips (dy). Equations need inverting.



Solution 4 (Method 1: Cylindrical Shells)

Setup: Vertical cuts (dx). Range $x \in [0, 1]$.

- **Height** $h(x)$: Top curve minus Bottom curve.

$$h(x) = \sqrt{x} - x^2$$

- **Radius** $r(x)$: Distance from cut at x to axis $x = -1$.

$$r(x) = x - (-1) = x + 1$$

Integral:

$$\begin{aligned} V &= \int_0^1 2\pi \underbrace{(x+1)}_{\text{radius}} \underbrace{(\sqrt{x} - x^2)}_{\text{height}} dx \\ &= 2\pi \int_0^1 (x^{3/2} - x^3 + x^{1/2} - x^2) dx \end{aligned}$$

Evaluate:

$$\begin{aligned} V &= 2\pi \left[\frac{2}{5}x^{5/2} - \frac{1}{4}x^4 + \frac{2}{3}x^{3/2} - \frac{1}{3}x^3 \right]_0^1 \\ &= 2\pi \left(\frac{2}{5} - \frac{1}{4} + \frac{2}{3} - \frac{1}{3} \right) = 2\pi \left(\frac{24 - 15 + 20}{60} \right) = \frac{29\pi}{30} \end{aligned}$$

Solution 4 (Method 2: Washers)

Setup: Horizontal cuts (dy). Range $y \in [0, 1]$. Need $x = f(y)$.

- **Right Curve (R):** $y = x^2 \implies x = \sqrt{y}$ (Farther from axis).

$$R(y) = \text{distance to } x = -1 = \sqrt{y} - (-1) = \sqrt{y} + 1$$

- **Left Curve (r):** $y = \sqrt{x} \implies x = y^2$ (Closer to axis).

$$r(y) = \text{distance to } x = -1 = y^2 - (-1) = y^2 + 1$$

Integral:

$$\begin{aligned} V &= \pi \int_0^1 ([R(y)]^2 - [r(y)]^2) dy \\ &= \pi \int_0^1 ((\sqrt{y} + 1)^2 - (y^2 + 1)^2) dy \\ &= \pi \int_0^1 (y + 2y^{1/2} - y^4 - 2y^2) dy \end{aligned}$$

Evaluate:

$$V = \pi \left[\frac{y^2}{2} + \frac{4}{3}y^{3/2} - \frac{y^5}{5} - \frac{2}{3}y^3 \right]_0^1 = \pi \left(\frac{1}{2} + \frac{4}{3} - \frac{1}{5} - \frac{2}{3} \right) = \frac{29\pi}{30}$$

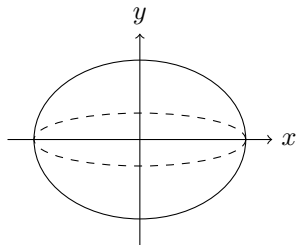
Problem 5: Volume of an Ellipsoid

Problem 5. Find the volume enclosed by the ellipsoid:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

Strategy (General Slicing):

- Fix x . Cross-section is an ellipse in yz -plane.
- Equation: $\frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 - \frac{x^2}{a^2}$.
- Area of ellipse: $\pi \cdot (\text{semi-axis}_1) \cdot (\text{semi-axis}_2)$.



Slicing along x-axis

Solution 5

1. Cross-sectional Area $A(x)$: Rearrange the equation for fixed x :

$$\frac{y^2}{b^2(1 - \frac{x^2}{a^2})} + \frac{z^2}{c^2(1 - \frac{x^2}{a^2})} = 1$$

Semi-axes are $b\sqrt{1 - \frac{x^2}{a^2}}$ and $c\sqrt{1 - \frac{x^2}{a^2}}$.

$$A(x) = \pi \left(b\sqrt{1 - \frac{x^2}{a^2}} \right) \left(c\sqrt{1 - \frac{x^2}{a^2}} \right) = \pi bc \left(1 - \frac{x^2}{a^2} \right)$$

2. Integrate:

$$\begin{aligned} V &= \int_{-a}^a A(x) dx = 2\pi bc \int_0^a \left(1 - \frac{x^2}{a^2} \right) dx \\ &= 2\pi bc \left[x - \frac{x^3}{3a^2} \right]_0^a \\ &= 2\pi bc \left(a - \frac{a}{3} \right) = \frac{4}{3}\pi abc \end{aligned}$$

Problem 6: Application of First MVT

Problem 6. Use the **First Mean Value Theorem for Integrals** to prove that:

$$\frac{1}{6} \leq \int_0^1 \frac{x^2}{1+x} dx \leq \frac{1}{3}$$

Recall: First Mean Value Theorem

If f is continuous and g is integrable with $g(x) \geq 0$ on $[a, b]$, then there exists $c \in [a, b]$ such that:

$$\int_a^b f(x)g(x) dx = f(c) \int_a^b g(x) dx$$

Hint: Choose $f(x)$ and $g(x)$ wisely. Which part is hard to integrate? Which part is strictly positive?

Solution 6

1. Identification: Let $I = \int_0^1 \frac{x^2}{1+x} dx$. We choose:

- $f(x) = \frac{1}{1+x}$ (Continuous on $[0, 1]$)
- $g(x) = x^2$ (Integrable and $g(x) \geq 0$ on $[0, 1]$)

2. Apply Theorem: By the First Mean Value Theorem, there exists some $c \in [0, 1]$ such that:

$$\int_0^1 \frac{x^2}{1+x} dx = f(c) \int_0^1 x^2 dx$$

3. Evaluate Components:

- $f(c) = \frac{1}{1+c}$
- $\int_0^1 x^2 dx = \left[\frac{x^3}{3} \right]_0^1 = \frac{1}{3}$

So, the integral is $I = \frac{1}{3(1+c)}$.

4. Bounding: Since $0 \leq c \leq 1$:

$$1 + 0 \leq 1 + c \leq 1 + 1 \implies \frac{1}{2} \leq \frac{1}{1+c} \leq 1$$

$$\frac{1}{3} \cdot \frac{1}{2} \leq \frac{1}{3(1+c)} \leq \frac{1}{3} \cdot 1 \implies \frac{1}{6} \leq I \leq \frac{1}{3}$$



Problem 6: The Wallis Inequality

Problem 6. Let $I_n = \int_0^{\pi/2} \sin^n x \, dx$ for $n \geq 0$.

(a) Use the property of $\sin x$ to show that the sequence I_n is monotonically decreasing:

$$I_{n+2} \leq I_{n+1} \leq I_n$$

(b) Use the **Reduction Formula** ($I_n = \frac{n-1}{n} I_{n-2}$) derived in Problem 3 to prove:

$$\frac{n}{n+1} \leq \frac{I_{n+1}}{I_n} \leq 1$$

And deduce the limit $\lim_{n \rightarrow \infty} \frac{I_{n+1}}{I_n}$.

Why this matters?

This inequality is the key to proving the famous **Wallis Product** for π :

$$\frac{\pi}{2} = \lim_{n \rightarrow \infty} \frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdot \frac{6}{5} \cdot \frac{6}{7} \cdots$$

Solution 6

1. Monotonicity of I_n : Since $0 \leq \sin x \leq 1$ for $x \in [0, \pi/2]$, we have:

$$\sin^{n+2} x \leq \sin^{n+1} x \leq \sin^n x$$

Integrating over $[0, \pi/2]$ gives:

$$I_{n+2} \leq I_{n+1} \leq I_n \quad \dots (*1)$$

2. Apply Reduction Formula: From Problem 3, we know $I_n = \frac{n-1}{n} I_{n-2}$.

Applying this to I_{n+2} , we get:

$$I_{n+2} = \frac{(n+2)-1}{n+2} I_n = \frac{n+1}{n+2} I_n$$

3. Construct the Sandwich Inequality: Substitute I_{n+2} back into (*1):

$$\frac{n+1}{n+2} I_n \leq I_{n+1} \leq I_n$$

Divide everything by I_n (since $I_n > 0$):

$$\frac{n+1}{n+2} \leq \frac{I_{n+1}}{I_n} \leq 1$$

Solution 6

4. Take the Limit: We now examine the limits of the lower and upper bounds as $n \rightarrow \infty$:

- **Lower Bound:**

$$\lim_{n \rightarrow \infty} \frac{n+1}{n+2} = \lim_{n \rightarrow \infty} \frac{1+1/n}{1+2/n} = 1$$

- **Upper Bound:**

$$\lim_{n \rightarrow \infty} 1 = 1$$

5. Conclusion: By the **Squeeze Theorem**, since the ratio is "squeezed" between two sequences that both converge to 1:

$$\lim_{n \rightarrow \infty} \frac{I_{n+1}}{I_n} = 1$$

THANK YOU!

